

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

2 DATA AND METHODOLOGY

We draw data from the O*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-